6 On the Earth, an object takes time T_E to fall from rest through a vertical distance h.

On the Moon, the same object takes time T_M to fall from rest through the same vertical distance h.

The ratio $\frac{\text{acceleration of free fall on the Earth}}{\text{acceleration of free fall on the Moon}}$ is equal to 6.

Air resistance is negligible for the object on the Earth and on the Moon.

What is the ratio $\frac{T_E}{T_M}$?

A $\frac{1}{6}$ **B** $\frac{1}{\sqrt{6}}$ **C** $\sqrt{6}$ **D** 6